

Section A

For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 At gas stations, there are 4 different grades of gasoline to choose from. The grade of the gasoline is dependent on the percentage by mass of octane (e.g. Grade 92 indicates 92% by mass of octane). Given that 1 kg of a certain grade of gasoline requires 2.39 m^3 of oxygen for complete combustion, what is the grade of the gasoline?
- A** 85 **B** 87 **C** 91 **D** 93

- 2 Zn reacts in a series of 3 reactions stepwise with VO_3^- ions to give V^{n+} . 3.9 g of Zn was required to react completely with 40.0 cm^3 of 1.0 mol dm^{-3} of KVO_3 .

What is value of n?

- A** 1 **B** 2 **C** 3 **D** 4

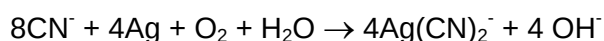
- 3 The first 8 successive ionisation energies, in kJ mol^{-1} , of an element X are given below.

1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
577	1820	2740	11600	12500	13500	14800	15900

What is the electronic configuration of X^{2+} ?

- A** $[\text{He}] 2s^1$ **B** $[\text{He}] 2s^2 2p^1$ **C** $[\text{Ne}] 3s^1$ **D** $[\text{Ne}] 3s^2 3p^1$

- 4 Aqueous solutions containing cyanide ions are often used to extract silver from their ores using the following reaction:



What is the amount of electrons involved in the reaction if 1 mole of silver was reacted?

- A** 0.25 mol **B** 0.5 mol **C** 0.75 mol **D** 1 mol

- 5 Which statement best describes the type(s) of bonding present in $(\text{NH}_4)_2\text{SO}_4$?

- A** ionic bonding
B ionic and covalent bonding
C ionic and hydrogen bonding
D ionic, covalent and hydrogen bonding

- 6 In which of the following pairs do the species **not** have similar shape?

- A** AlCl_3 and CO_3^{2-} **B** NH_4^+ and CH_4
C NH_3 and H_3O^+ **D** SeF_4 and SO_4^{2-}

- 7 Which of the following best explains why sodium oxide has a lower melting point than aluminium oxide?
- A** As Al^{3+} ion has a smaller radius than Na^+ , aluminium oxide has a less exothermic lattice energy.
- B** As Al^{3+} has a higher charge and smaller radius than Na^+ , aluminium oxide has a stronger ionic bond than sodium oxide.
- C** As Al^{3+} ion has a larger charge density than Na^+ , aluminium oxide is an ionic compound with covalent character, whereas sodium oxide is an ionic compound.
- D** As Al^{3+} has a higher charge than Na^+ , there are more electrons involved in bonding and hence, the aluminium oxide has a stronger ionic bond than sodium oxide.

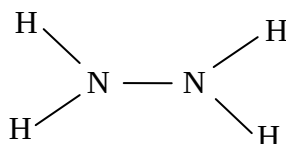
- 8 A solid **G** has the following physical properties.

- o It is insoluble in water.
- o It melts at 1310°C .
- o It conducts electricity in molten state.
- o It is hard and brittle.

What is the likely structure of **G**?

- | | |
|--------------------------------------|--|
| A an ionic crystal lattice | B an atomic crystal lattice |
| C a molecular crystal lattice | D a giant molecular crystal lattice |

- 9 Hydrazine, N_2H_4 , is useful as a rocket fuel. It has properties that are similar to those of ammonia, NH_3 .



Which of the following describes correctly the property of hydrazine as compared to ammonia?

- A** Hydrazine has a trigonal planar shape like ammonia.
- B** Hydrazine can conduct electricity in its liquid form but ammonia cannot.
- C** Hydrazine is more acidic than ammonia as there are more hydrogen atoms present.
- D** Hydrazine is more soluble in water as it can form more hydrogen bonds with water than ammonia.

- 10 0.0500 mol of solid $\text{Ba}(\text{OH})_2$ was added to 1 dm^3 of $0.100 \text{ mol dm}^{-3}$ NaOH (aq). What is the resultant pH of the solution?

- | | | | |
|---------------|---------------|---------------|---------------|
| A 13.0 | B 13.2 | C 13.3 | D 13.5 |
|---------------|---------------|---------------|---------------|

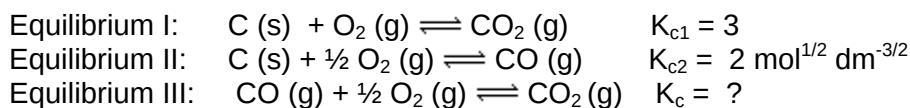
- 11 Bleaching solutions are manufactured by dissolving chlorine gas in sodium hydroxide solution to give the following reaction.



Users are warned not to mix the bleach with other cleaning solutions to prevent evolution of hazardous chlorine gas. Which of the following actions will lead to liberation of chlorine gas?

- A Addition of water to bleach.
- B Mixing of alkali with bleach.
- C Subjecting bleach to high pressure.
- D Shaking bleach with table salt (NaCl).

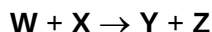
- 12 Given that,



Using K_{c1} and K_{c2} , what is the numerical K_c value for the Equilibrium III?

- A $\frac{\sqrt{2}}{3}$
- B $\frac{2}{3}$
- C $\frac{\sqrt{3}}{2}$
- D $\frac{3}{2}$

- 13 If the reaction,



is described as being first order with respect to **W**, which of the following statements can be deduced?

- A **W** is acting as a catalyst.
- B The half-life of the reaction is constant.
- C The rate of reaction is dependent directly on the concentration of **W**.
- D All molecules of **W** possess sufficient activation energy to react.

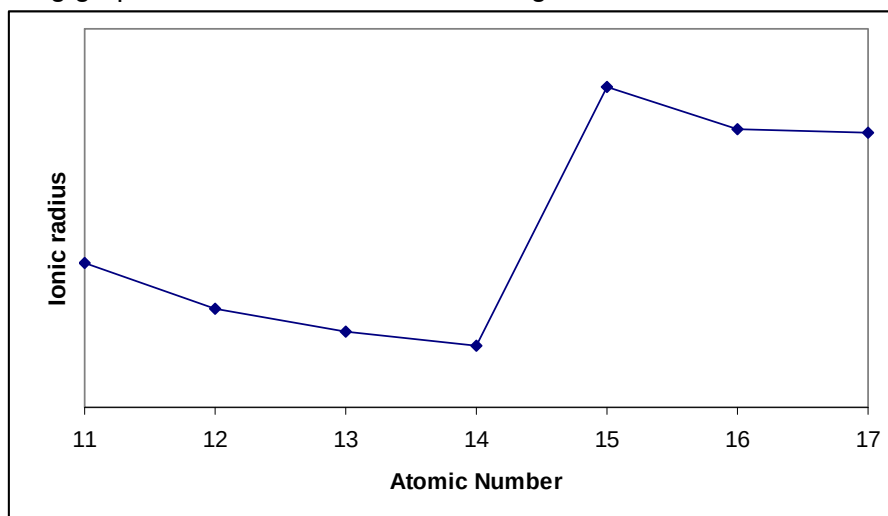
- 14 Which statement about the chemical properties of the oxides in the third period of the Periodic Table is true?

- A Al_2O_3 is soluble in both KOH and HCl.
- B P_2O_5 reacts with water to give OH^- ions.
- C SiO_2 forms a solution of pH 6 when dissolved in water at room temperature.
- D Na_2O and MgO can be mixed in water to give an approximately neutral solution.

- 15 Which of the following correctly represents both the lattice energy of potassium fluoride and the bond energy of iodine?

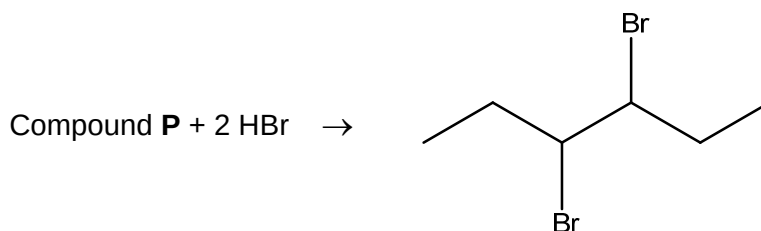
- | | Lattice energy | Bond energy |
|---|---|---|
| A | $\text{KF (s)} \rightarrow \text{K}^+(\text{g}) + \text{F}^-(\text{g})$ | $\text{I}_2 (\text{s}) \rightarrow 2 \text{I} (\text{g})$ |
| B | $\text{K}^+(\text{g}) + \text{F}^-(\text{g}) \rightarrow \text{KF (s)}$ | $\text{I}_2 (\text{g}) \rightarrow 2 \text{I} (\text{g})$ |
| C | $\text{KF (s)} \rightarrow \text{K}^+(\text{aq}) + \text{F}^-(\text{aq})$ | $\text{I}_2 (\text{s}) \rightarrow 2 \text{I} (\text{g})$ |
| D | $\text{K}^+(\text{aq}) + \text{F}^-(\text{aq}) \rightarrow \text{KF (s)}$ | $\text{I}_2 (\text{g}) \rightarrow 2 \text{I} (\text{g})$ |

- 16 The following graph shows how ionic radius changes across Period 3 for 7 elements.



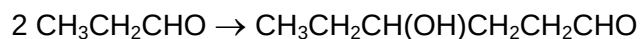
Which of the following correctly explains the trend?

- A** The sharp increase in radius between the 4th and 5th element is due to an increase in the number of principal quantum shells.
- B** There is a general decrease in ionic radius for the first 4 elements in Period 3 due to decreasing shielding effect.
- C** There is a general decrease in ionic radius for the last 3 elements due to decreasing proton to electron ratio.
- D** The first 4 elements form anions hence have lower ionic radii than the last 3 elements which form cations.
- 17 Compound **P** reacts with 2 mol of HBr at room temperature in the following reaction:



Given that there are no adjacent carbon-carbon double bonds in **P**, what is the total number of cis-trans isomers that **P** can have?

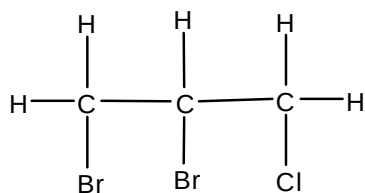
- A** 1 **B** 2 **C** 3 **D** 4
- 18 Propanal, $\text{CH}_3\text{CH}_2\text{CHO}$, can react with itself to form a compound known as aldol.



Which of the following reagents and conditions can distinguish between the ethanal and aldol?

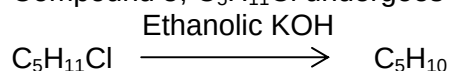
- A** Acidified potassium manganate (VII), heat
- B** 2,4 dinitrophenylhydrazine, room temperature
- C** Phosphorous pentachloride, room temperature
- D** Hydrogen cyanide with trace amount of sodium cyanide, $10^\circ\text{C} - 20^\circ\text{C}$

- 19 1,2-Dibromo-3-chloropropane (DBCP) has been used in the control of earthworms in agricultural land. The structure of DBCP is shown below.



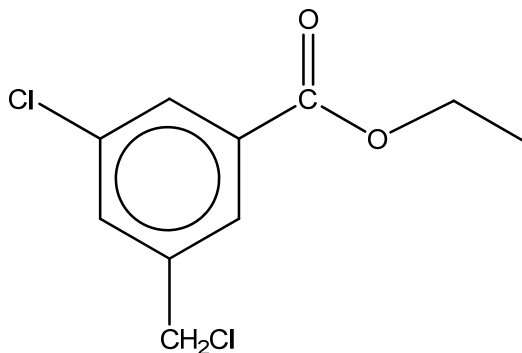
Which of the following will lead to highest yield of DBCP?

- A $\text{CH}_2=\text{CHCH}_2\text{Cl} + \text{Br}_2 \rightarrow \text{DBCP}$
 B $\text{CH}_2=\text{CHCHBr}_2 + \text{HCl} \rightarrow \text{DBCP}$
 C $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + 2\text{Br}_2 \rightarrow \text{DBCP} + 2\text{HBr}$
 D $\text{CH}_3\text{CHBrCH}_2\text{Br} + \text{BrCl} \rightarrow \text{DBCP} + \text{HCl}$
- 20 Compound J, $\text{C}_5\text{H}_{11}\text{Cl}$ undergoes the following reaction.



Which of the following **cannot** be Compound J?

- A 1-chloropentane
 B 2-chloropentane
 C 2-chloro-3-methylbutane
 D 1-chloro-2,2-dimethylpropane
- 21 Compound M, as shown below, was heated with alkaline iodine and acidified with aqueous hydrochloric acid thereafter.

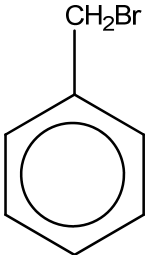
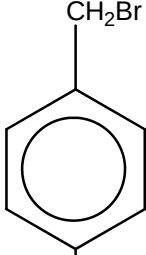
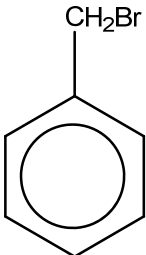
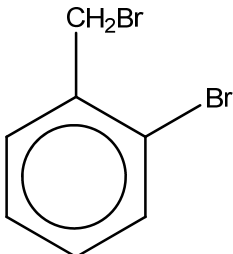
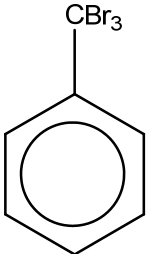
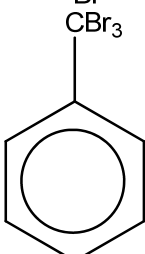
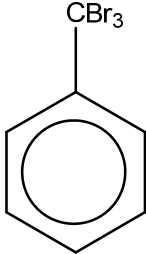
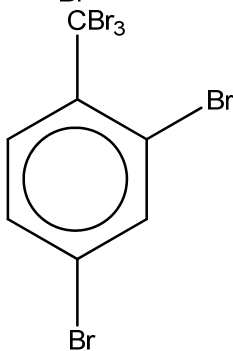


What is the total number of organic products obtained?

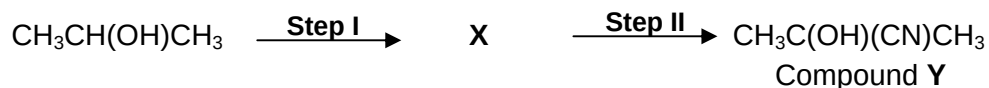
- A 1 B 2 C 3 D 4
- 22 Chlorofluorocarbons (CFCs) are known to cause environmental problems. Which of the following identifies the main cause for the environmental unfriendliness of CFCs?
- A Breaking of carbon –hydrogen bond
 B Breaking of carbon – halogen bond
 C Breaking of carbon – carbon bond
 D Breaking of carbon – oxygen bond

- 23 Methylbenzene and bromine, in the ratio of 1:6 were mixed and left under the sun and compound **Q** was isolated. After which, iron fillings were added to the mixture at room temperature and Compound **R** was identified to be the final product.

Which of the following is likely to be Compounds **Q** and **R**?

	Compound Q	Compound R
A		
B		
C		
D		

- 24 The reaction scheme below shows the formation of compound Y:

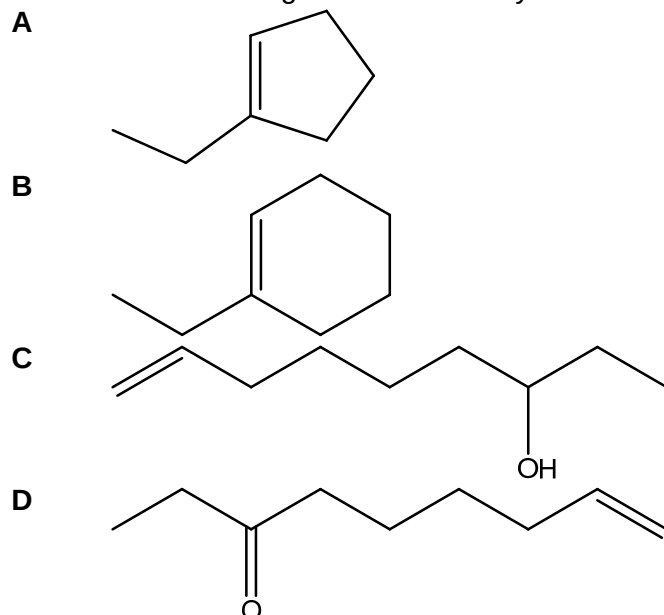


Which of the following suggests the reagents and conditions that will give the best yield of Compound Y?

	Step I	Step II
A	Br_2 (l), UV light	ethanolic KCN
B	Br_2 (l), in the dark	HCN in trace amount of NaCN, 10-20°C
C	Acidified $\text{K}_2\text{Cr}_2\text{O}_7$, distill	ethanolic KCN
D	Acidified $\text{K}_2\text{Cr}_2\text{O}_7$, heat under reflux	HCN in trace amount of NaCN, 10-20°C

- 25 A hydrocarbon, on heating with acidified concentrated KMnO_4 gives $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$.

Which of the following **cannot** be the hydrocarbon?



Section B

From questions **26** to **30**, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it useful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 are correct	2 and 3 are correct	Only 1 is correct

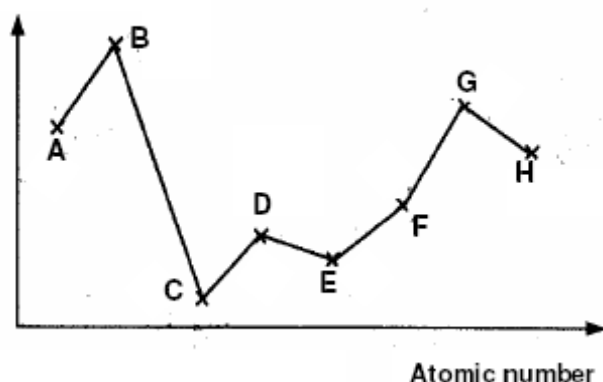
No other combination of statements is used as a correct response.

26 Which of the following species has/have unpaired electrons?

- 1** Cl
- 2** N^{3-}
- 3** Ne

27 The following graph shows the first ionisation energy of 8 consecutive elements, from **A** to **H** in the Periodic Table with atomic number between 3 to 20.

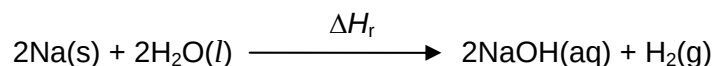
Ionisation energy



Which of the following is/are true?

- 1** The chloride of **D** conducts electricity in the molten state.
- 2** **A** reacts with **C** to form a compound with giant ionic lattice structure.
- 3** The melting points of **A** to **H** also follow the trend as shown by the graph.

- 28 The enthalpy change of reaction, ΔH_r , between sodium and water (in excess) to produce sodium hydroxide and hydrogen gas can be measured in the laboratory.



Other than temperature change of the solution, what information is needed to calculate a value for the enthalpy change of this reaction?

- 1 Mass of water
 - 2 Mass of sodium
 - 3 Pressure
- 29 Reaction of compound **K** with excess concentrated H_2SO_4 at 170°C gives 2-methylpropene. Compound **K** was distilled with acidified potassium dichromate(VI) and the resulting product was added into a test-tube containing diammine silver(I) ions. A silver mirror was obtained.

What are the types of reaction involved above?

- 1 Elimination
 - 2 Oxidation
 - 3 Substitution
- 30 Which of the following reactions *could* have the same common intermediate?
- 1 $\text{CH}_3\text{CH}=\text{CH}_2 \rightarrow \text{intermediate} \rightarrow \text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_3$
 - 2 $\text{CH}_3\text{CH}=\text{CH}_2 \rightarrow \text{intermediate} \rightarrow \text{CH}_3\text{COCH}_3$
 - 3 $\text{CH}_3\text{CO}_2\text{CH}(\text{CH}_3)_2 \rightarrow \text{intermediate} \rightarrow \text{CH}_3\text{CHBrCH}_3$